b) $y' + y\cos(x) = \operatorname{sen}(x)\cos(x)$

$$y' + P(x)y = Q(x)$$

$$r(x) = e^{\int P(x)dx}$$

$$y' \cdot r(x) + r(x) \cdot P(x) \cdot y = Q(x) \cdot r(x)$$

$$r'(x) = e^{\int P(x)dx} \cdot P(x) = r(x) \cdot P(x)$$

$$y' \cdot r(x) + r'(x)\, y = Q(x) \cdot r(x)$$

$$\int (y \cdot r(x))' dx = \int Q(x) \cdot r(x)\, dx$$

$$y = \frac{\int Q(x) \cdot r(x)\, dx}{r(x)}$$

b) $y' + y\cos(x) = \operatorname{sen}(x)\cos(x)$

$$P(x) = \cos x$$

$$r(x) = e^{\int \cos x\, dx} = e^{\operatorname{sen} x + k} = e^{\operatorname{sen} x} \cdot e^{k}$$

$$= A \cdot e^{\operatorname{sen} x}$$

$$y = \frac{\int \operatorname{sen} x \cdot \cos x \cdot \cancel{A} \cdot e^{\operatorname{sen} x} \, dx}{\cancel{A} \cdot e^{\operatorname{sen} x}}$$

$$y = \frac{\int \operatorname{sen} x \cdot \cos x \cdot e^{\operatorname{sen} x} \, dx}{e^{\operatorname{sen} x}}$$

$$\operatorname{sen} x = u$$
$$\cos x \, dx = du$$

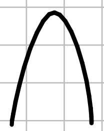
$$\int \operatorname{sen} x \cos x \, e^{\operatorname{sen} x} \, dx = \int u \cdot e^{u} \cdot du = e^{u} \cdot (u - 1) + c$$

$$(196) \quad \int x e^{ax} \, dx = \frac{e^{ax}}{a}\left(x - \frac{1}{a}\right)$$

$$= e^{\operatorname{sen} x}(\operatorname{sen} x - 1) + c$$

$$y = \frac{e^{\operatorname{sen} x}(\operatorname{sen} x - 1) + c}{e^{\operatorname{sen} x}} = \operatorname{sen} x - 1 + \frac{c}{e^{\operatorname{sen} x}}$$

Vert. $(0,2,4)$



$$x^2 + (y-2)^2 = -z+4$$

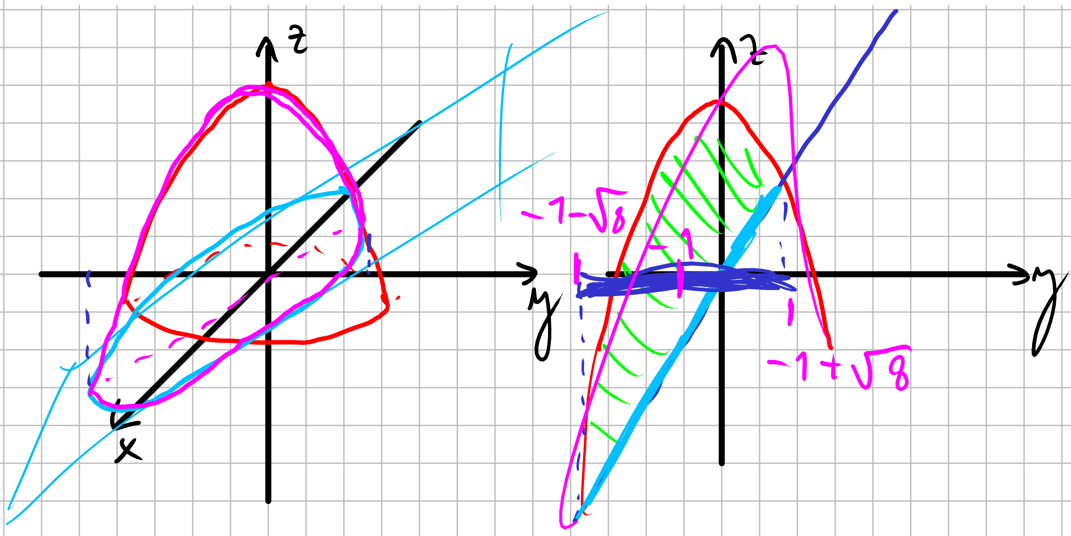
$$\Sigma : x^2 + (y-2)^2 = -(z-4) \quad , \quad z \geq 2$$

$$\Sigma_1 : x^2 + (y-2)^2 = -(z-4)$$

$$\Sigma_2 : z = 2$$

$$S : \Sigma_1 \cup \Sigma_2 \cap x^2 + (y-2)^2 \leq 2$$

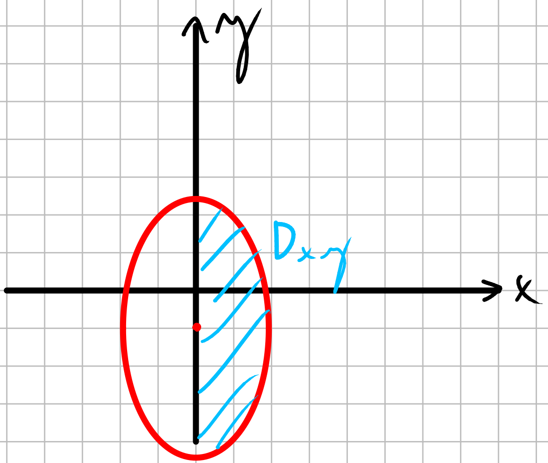
4. **Calcule** el volumen del cuerpo D definido por $2x^2+y^2+z\le 7$, $z\ge 2y$, $x\ge 0$.



$$2x^2+y^2+2y+1-1=7$$

$$2x^2+(y+1)^2=8$$

$$\frac{x^2}{4}+\frac{(y+1)^2}{8}=1$$



$$\iiint_D 1\,dV=\iint_{D_{xy}}\int_{2y}^{7-2x^2-y^2} 1\,dz\,dx\,dy$$

$$=\iint_{D_{xy}}\left(7-2x^2-y^2-2y\right)dx\,dy$$

$$=\iint_{D_{xy}} 7-2x^2-(y^2+2y+1-1)\,dx\,dy$$

$$=\iint_{D_{xy}}\left(7-2x^2-(y+1)^2+1\right)dx\,dy$$

$$=\iint_{D_{xy}}\left(8-2x^2-(y+1)^2\right)dx\,dy$$

$$x=2r\cos\theta$$

$$y=\sqrt{8}\,r\,\mathrm{sen}\,\theta-1$$

$$|J|=2\sqrt{8}\,r=2\sqrt{4\cdot 2}\,r=4\sqrt{2}\,r$$

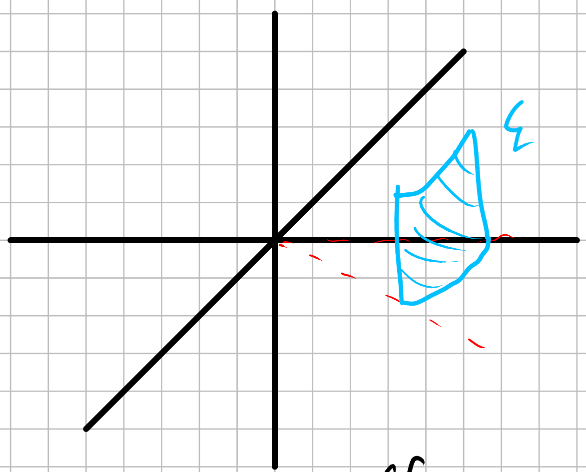
$$= \iint_{D_{xy}} \left(8 - 2x^2 - (y+1)^2\right) dx$$

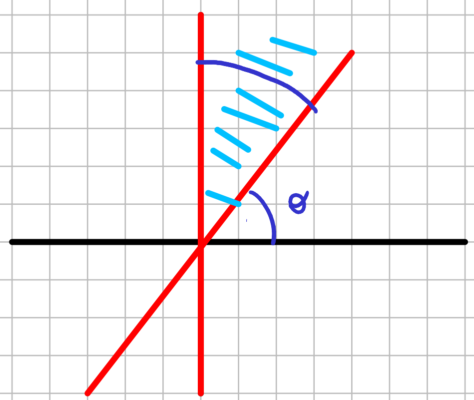
$$= \int_0^1 \int_{-\pi/2}^{\pi/2} \left(8 - 8r^2\cos^2\theta - 8r^2\sin^2\theta\right) \cdot 4\sqrt{2}\, r \, d\theta \, dr$$

$$= \int_0^1 \int_{-\pi/2}^{\pi/2} \left(8 - 8r^2\right) 4\sqrt{2} r \, d\theta \, dr$$

$$= \pi \cdot 4 \cdot \sqrt{2} \int_0^1 \left(8 - 8r^2\right) \cdot r \, dr$$

$$x^2 + y^2 = 4 \,, \quad 0 \le z \le 2 \,, \quad 0 \le x \le \frac{y}{\sqrt{3}}$$





$y = \sqrt{3} x$

$\operatorname{tg}\theta = \frac{y}{x} = \sqrt{3}$

$\operatorname{tg}\theta = \sqrt{3}$

$\theta = \operatorname{arctg} \sqrt{3}$

$\theta = \frac{\pi}{3}$

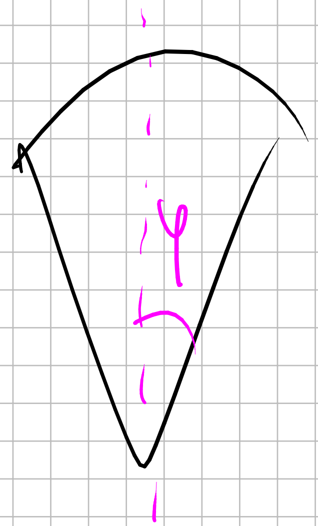
$$\text{Area } \Sigma = \iint_\Sigma 1 \, dS = \iint_D \|\vec{\sigma}'_u \times \vec{\sigma}'_v\| \, du \, dv$$

$$\vec{\sigma}(\theta, z) = (2\cos\theta, 2\sin\theta, z)$$

$\theta \in \left[\frac{\pi}{3}, \frac{\pi}{2}\right]$

$z \in [0, 2]$

$$\|\vec{\sigma}'_\theta \times \vec{\sigma}'_z\| = 2$$



$3 \; z = \sqrt{x^2 + y^2}$

$x^2 + y^2 + z^2 = 1$

$$\int_0^2 \int_{\pi/3}^{\pi/2} 2 \, d\theta \, dz$$

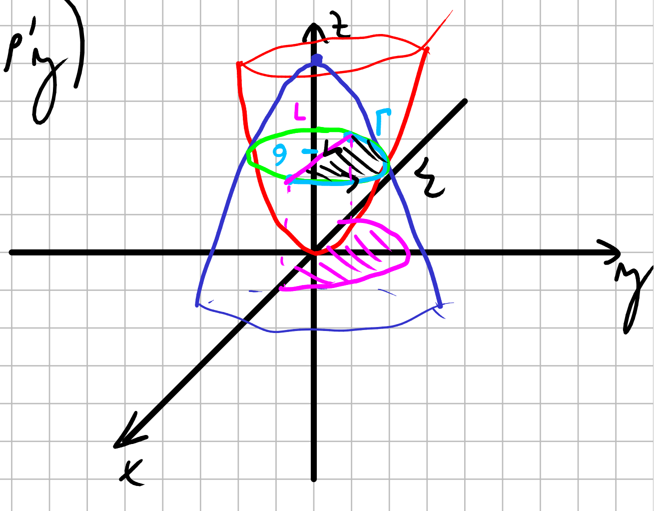
- **Ejercicio 5.** Calcule la circulación de $\vec{f}(x,y,z) = (x^2z - y, x + y^3e^y, z^2)$ a lo largo de la porción de la curva $C = \Sigma_1 \cap \Sigma_2$ contenida en el semiespacio $y \geq 0$, donde

$$\Sigma_1 : x^2 + y^2 - z = 0 \quad \wedge \quad \Sigma_2 : x^2 + y^2 + z = 18.$$

Indique en un gráfico el sentido de recorrido de la curva que considera en el cálculo.

$\operatorname{rot} \vec{F} = (R'_y - Q'_z, \; P'_z - R'_x, \; Q'_x - P'_y)$

$= (0, x^2, 2)$



$$\oint_{\partial\Sigma} \vec{F} \cdot \vec{dl} = \iint_\Sigma \nabla \times \vec{F} \cdot \vec{ds}$$

$$\int_\Gamma \vec{F} \cdot \vec{dl} + \int_L \vec{F} \cdot \vec{dl} = \iint_\Sigma \nabla \times \vec{F} \cdot \vec{ds}$$

$$\Sigma_1 : x^2 + y^2 - z = 0 \quad \wedge \quad \Sigma_2 : x^2 + y^2 + z = 18.$$

$z = x^2 + y^2$

$z = 18 - x^2 - y^2$

$x^2 + y^2 = 18 - x^2 - y^2$

$z = 9$

$x^2 + y^2 = 9$

$$\iint_{\Sigma} \nabla\times\vec{F}\cdot\vec{dS} = \begin{cases} \iint_{\Sigma} \nabla\times\vec{F}\cdot\check{n}\, dS \\ \iint_{D} \nabla\times\vec{F}(\underbrace{\vec{\sigma}(u,v)}_{(x,y,z)})\cdot\vec{\sigma}'_u\times\vec{\sigma}'_v\, du\, dv \end{cases}$$

$$\iint_{\Sigma} (0, x^2, 2)\cdot(0,0,1)\, dS = \iint_{\Sigma} 2\, dS = 2\underbrace{\text{Area}\,\Sigma}_{\frac{9\pi}{2}} = 9\pi$$

$$L: \vec{\rho}(t) = (t, 0, 9) \quad t\in[-3,3]$$

$$\vec{\rho}'(t) = (1,0,0)$$

$$\int_{-3}^{3} (9t^2, \dots, \dots)(1,0,0)\, dt = \int_{-3}^{3} 9t^2\, dt$$

$$= 2\int_0^3 9t^2\, dt = 2\cdot 3t^3\Big|_0^3$$

$$= 162$$

$$\int_{\Gamma} \vec{F}\cdot\vec{dl} + 162 = 9\pi$$

$$\int_{\Gamma} \vec{F}\cdot\vec{dl} = 9\pi - 162$$

$$z = 18 - x^2 - y^2$$ ojito ☺

$$z = \sqrt{x^2+y^2}$$

$z>0$ $\quad z^2 = x^2+y^2$

$$z = 18 - z^2$$

un cubo H de arista 2 sea 16, cuando ∂H se orienta tomando la normal

- **Ejercicio 5.** Sea $\vec{f} = \nabla\varphi + \vec{g}$ tal que $\varphi \in C^2(\mathbb{R}^3)$ y $\vec{g}$ tiene componentes en $C^1(\mathbb{R}^3)$. Calcule la circulación $\vec{f}$ a lo largo de la curva $\vec{X} = (1+2\cos(t), 2\,\mathrm{sen}(t), 1+\mathrm{sen}(t))$, $t \in [0,\pi]$, con la orientación dada por la parametrización, sabiendo que:

  1. $\varphi(x,0,1) = x^2$; $\qquad \varphi(3,0,1)=9$
  2. $\vec{g}(x,0,1) = (1,x,2)$; $\qquad \varphi(-1,0,1)=1$
  3. $\nabla \times \vec{g}(x,y,z) = (x, 2z-y, 0)$.

$0 \le y \le 2$

$1 \le z \le 2$

$\vec{x}(0) = (3,0,1)$

$\vec{x}(\frac{\pi}{2}) = (1,2,2)$

$\vec{x}(\pi) = (-1,0,1)$

$$\nabla\times\vec{F} = \underbrace{\cancel{\nabla\times\nabla\varphi}}_{\bar{0}\ \ C^2} + \nabla\times\vec{g} = (x, 2z-y, 0)$$

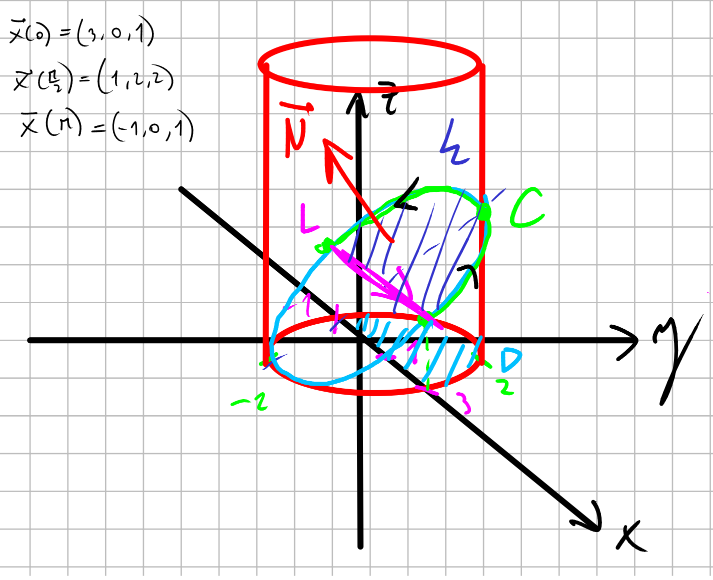
$x = 1+2\cos t \longrightarrow \left(\frac{x-1}{2}\right)^2 = \cos^2 t$

$y = 2\,\mathrm{sen}\, t \longrightarrow \left(\frac{y}{2}\right)^2 = \mathrm{sen}^2 t$

$z = 1+\mathrm{sen}\, t$

$$\frac{(x-1)^2}{4} + \frac{y^2}{4} = 1$$

$$(x-1)^2 + y^2 = 4$$



$$z = 1 + \frac{y}{2}$$

$$\int_{L} \vec{F}\cdot\vec{dl} + \int_{C} \vec{F}\cdot\vec{dl} = \iint_{\Sigma} \nabla\times\vec{F}\cdot\vec{dS}$$

$$\vec{\sigma}(x,y) = (x, y, 1+\tfrac{y}{2}) \quad (x,y)\in D$$

$$\vec{\sigma}'_x = (1, 0, 0)$$

$$\times$$

$$\vec{\sigma}'_y = (0, 1, 1/2)$$

$$(0, -\tfrac{1}{2}, 1)$$

$$z = 1 + \tfrac{y}{2}$$

$$2z = 2 + y$$

$$2z - y = 2$$

$$\iint_D (x, 2, 0)(0, -\tfrac{1}{2}, 1)\, dx\, dy$$

$$\iint_D (-1)\, dx\, dy = -\text{Area } D$$

$$= -2\pi$$

$$\vec{\gamma}(t) = (t, 0, 1) \quad t\in[-1,3]$$

$$\vec{\gamma}'(t) = (1, 0, 0)$$

$$\int_{L\searrow} \vec{F}\cdot d\vec{l} = \int_{L\searrow} \nabla\varphi\, d\vec{l} + \int_{L\searrow} \vec{g}\cdot d\vec{l}$$

$$= \varphi(3,0,1) - \varphi(-1,0,1) + \int_{-1}^{3} (1, t, 2)\cdot(1,0,0)\, dt$$

$$= 8 + \int_{-1}^{3} 1\, dt = 12$$

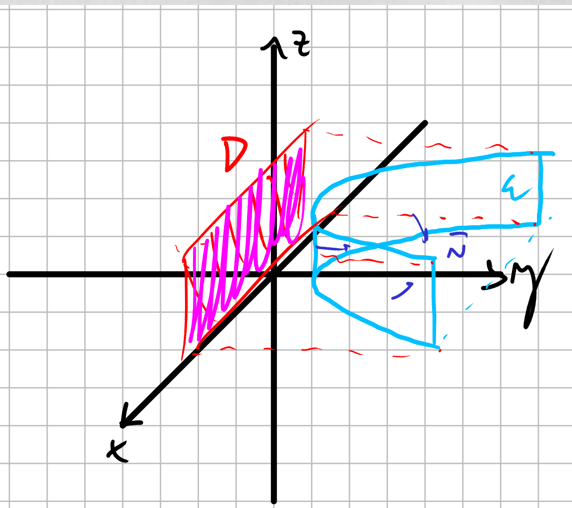
$$\int_{C\circlearrowleft} \vec{F}\cdot d\vec{l} + 12 = -2\pi$$

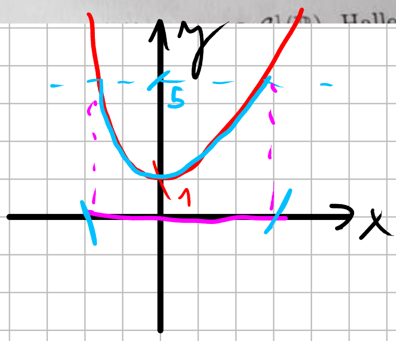
$$\int_{C\circlearrowleft} \vec{F}\cdot d\vec{l} = -2\pi - 12$$

- **Ejercicio 3.** Sea Σ la superficie definida por las condiciones $y - x^2 = 1$, $y \leq 5$ y $0 \leq z \leq 2$ orientada de modo que el versor normal en $(0,1,0)$ sea $\vec{j}$.

Calcule $\iint_\Sigma \vec{f} \cdot \vec{n}\, d\sigma$ con $\vec{f}(x,y,z) = (x, y, ze^x)$.

$\cup \to (0,1,0)$





$5 - x^2 = 1$

$x^2 = 4$

$x = \pm 2$

$\bar{\sigma}(x,z) = (x, 1+x^2, z) \quad (x,z) \in D$

$\bar{\sigma}'_x = (1, 2x, 0)$

$\times \quad \bar{\sigma}'_z = (0,0,1)$

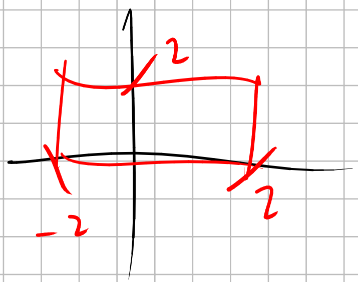
$(2x, -1, 0)$ ✗

Uso $(-2x, 1, 0)$

$$\iint_\Sigma \bar{F} \cdot d\bar{S} = \iint_D \bar{F}(\bar{\sigma}(u,v)) \cdot \bar{\sigma}'_u \times \bar{\sigma}'_v \, du\, dv$$

$$\iint_D (x, 1+x^2, \ldots)(-2x, 1, 0)\, dx\, dz = \iint_D (-2x^2 + 1 + x^2)\, dx\, dz$$

$$= \iint_D (1 - x^2)\, dx\, dz$$



$$= \int_{-2}^{2} \int_0^2 (1 - x^2)\, dz\, dx$$

$$= 2 \int_0^2 \int_0^2 (1 - x^2)\, dz\, dx$$

... un gráfico de H. $x-4 \leq z \leq 4-x$.

- **Ejercicio 2.** Sea D la región encerrada entre las curvas de ecuaciones $y^2 - 2y - x = 0$ y $x = 3$. Determine el valor de $a \in \mathbb{R}$ para que

$$\oint_{\partial D^+} 4ay\,dx + a^2 x\,dy$$

sea mínima y halle ese valor extremo.

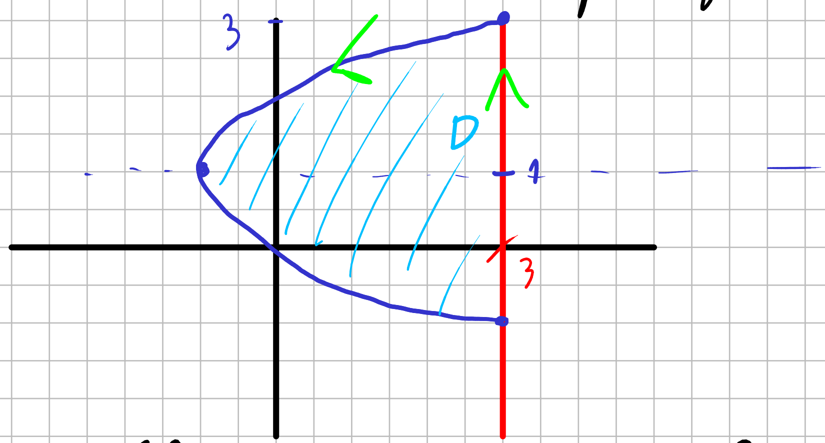
$\bar{F} = (P, Q) = (4ay, a^2 x)$

$x = y^2 - 2y \qquad x = 3$

$Q'_x - P'_y = a^2 - 4a$

$3 = y^2 - 2y$

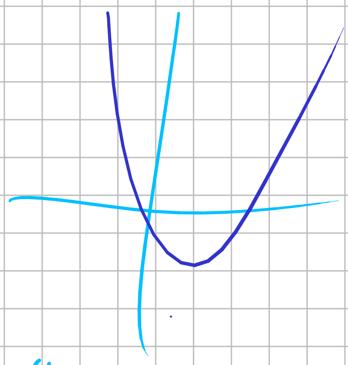
$y^2 - 2y - 3 = 0 \to y = 3, \; y = -1$



$$\oint_{\partial D^+} \bar{F} \cdot \overline{dl} = \iint_D (Q'_x - P'_y)\,dx\,dy = \iint_D (a^2 - 4a)\,dx\,dy$$

$$(a^2 - 4a) \iint_D 1\,dx\,dy$$

$g(a) =$



$\underbrace{(a^2 - 4a)}_{g(a)} \cdot \text{Area } D$

$g'(a) = 2a - 4 = 0$

$a = 2 \to$ MINIMIZA LA CIRC.

$$\oint_{\partial D^+} \bar{F} \cdot \overline{dl} = -4 \cdot \iint_D 1\,dx\,dy = -4 \cdot \int_{-1}^{3} \int_{y^2 - 2y}^{3} 1\,dx\,dy = -\frac{128}{3}$$